Classical & Keynesian Economics

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EX-ANTE AND EX-POST

Ex-ante means planned or intended or expected. For example, ex-ante investment means investment panned to be made during the year.

Ex-post means actual or realized. For example, ex-post investment means actual.

All variables in the theory of income determination are ex-ante variables.

AGGREGATE SUPPLY

Aggregate supply is the total volume goods and services the economy planned to be produced by all production units in the economy during a given period of time. The value of this output equals the cost planned to be incurred on producing this output. The cost includes factor payments such as wages, rents, interests, profits etc which in turn forms factor income. Factor incomes are either consumed or saved.

Thus, Aggregate Supply may be summarised as:

AGGREGATE (or TOTAL) SUPPLY= TOTAL PRODUCT = TOTAL (FACTOR) INCOME= CONSUMPTION EXPENDITURE + SAVING

AGGREGATE DEMAND

Aggregate demand is the total demand for final goods and services that the economy as a whole plan to buy at a given level of income time during a given period of time. Aggregate demand equals total (planned) expenditure for final goods and services (consumption and investment goods).

The component of Aggregate Demand in an open economy are

- **Private Consumption Expenditure (C)**
- **Private Investment Expenditure (I)**
- **Government Expenditure (G)**
- **Net Export i.e Export (X)– Import (M)**

Thus,
Aggregate Demand= Aggregate Expenditure= C + I + G + (X – M)

SOME NOTIONS ABOUT EMPLOYMENT AND UNEMPLOYMENT

- **Employment :** **Employment of a factor refers to its use in the process of production. However, the term 'employment' has been synonymously treated with employment of labour or workers. Worker is said to be employed when he is engaged in act of production.**

- **Unemployment:** **Unemployment** (or **joblessness**) **occurs when people are without work and actively seeking work.**

- **Voluntary Unemployment:** **Voluntary unemployment exists when people have chosen not to work because the wage at which they want to work is higher than the prevailing wage.**

- **Involuntary unemployment:** **Involuntary unemployment occurs when a person is willing to work at the prevailing wage yet is unemployed.**

- **Full employment:** **Full employment in a very simple sense may mean that the total available supply of labour is completely absorbed in gainful employment. Full-employment may mean absence of involuntary unemployment.**

Classical Economics: Assumption of Full-Employment

The entire economic premise of the classical economists was based on the assumption of full-employment of labour and other economic resources.

Classical economists argued that in the long-run under perfect competition a free capitalist laissez-faire economy would automatically tend to move towards full-employment (absence of involuntary unemployment) and unemployment would be voluntary.

Say's law of market lies at the centre of the classical notion of full employment. J B say, a French economist of the 19th century introduced the theory of market which states that "supply creates it demand".

Since supply creates its own demand there is no possibility of any general overproduction or deficiency in aggregate demand. Aggregate supply always equal aggregate demand.

When there is no general over-production, there is no general unemployment and free economy automatically attains equilibrium at full-employment level in the long-run. There is built-in-stability in an laissez-faire economy.

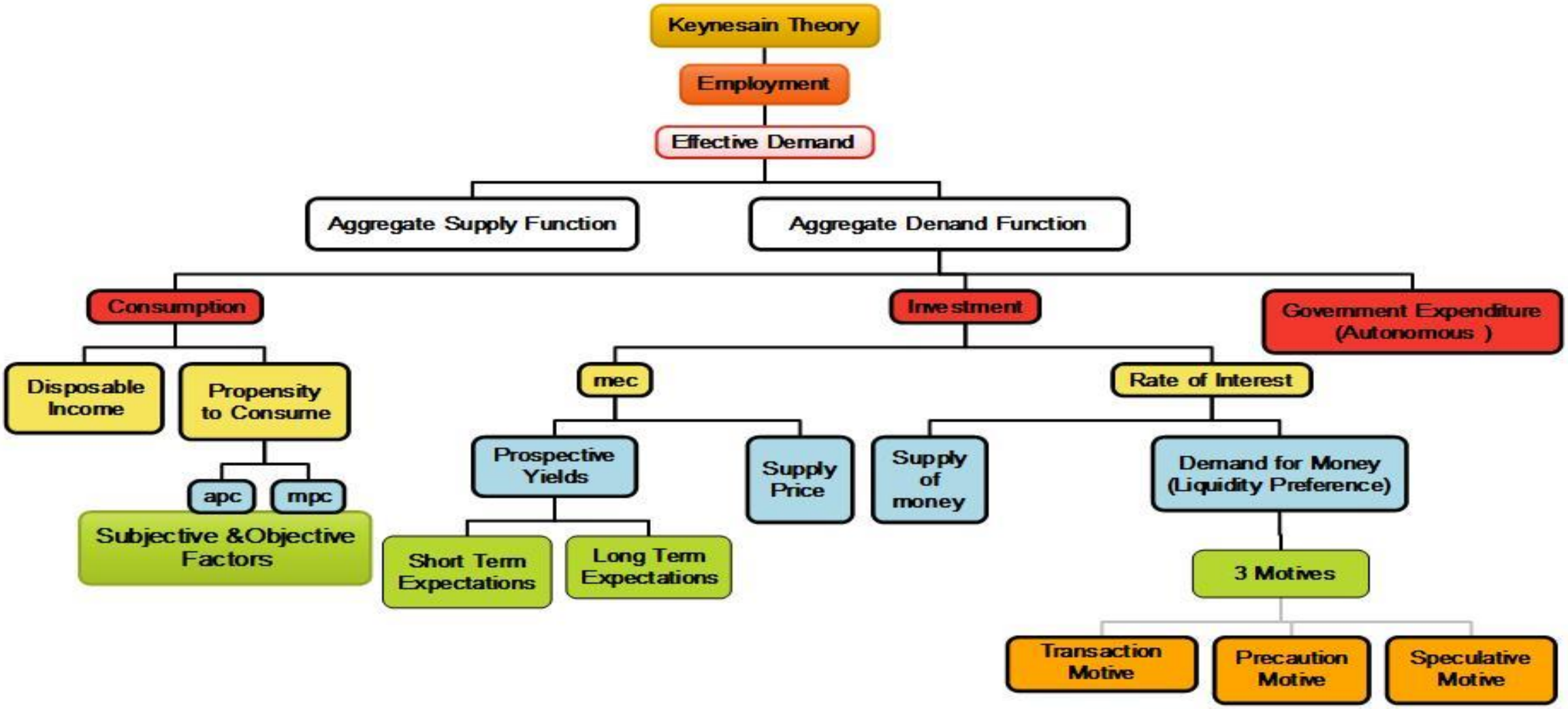
KEYNES' CRITICISM AGAINST CLASSICAL THEORY

Keynesian economics is the outcome of J.M Keynes' disagreement with the classicists. Keynes in his masterpiece 'The General Theory of Employment, Interest and Money' laid a frontal attack on the doctrine of classical economics.

- **Keynes considered the fundamental classical assumptions of full-employment equilibrium condition as rare and unrealistic and phenomenon. According to him, there is possibility of equilibrium at less than full-employment (underemployment) as a normal phenomenon.**

- **Keynes opposed the classical insistence on long-term equilibrium; instead he attached greater importance to short-term equilibrium. According to him, in the long run we are all dead.**

- **Classical economics rests on Say's law of market which blindly assures that supply always creates its own demand and affirmed impossibility of general over production and disequilibrium in the economy. Keynes totally disagreed with this and stress the possibility of supply exceeding demand, causing disequilibrium in the economy.**


Keynesain Theory
Employment
Effective Demand
Aggregate Supply Function
Aggregate Demand Function
Consumption
Investment
Government Expenditure (Autonomous)
Disposable Income
Propensity to Consume
apc
mpc
Subjective &Objective Factors
mec
Prospective Yields
Short Term Expectations
Long Term Expectations
Supply Price
Rate of Interest
Supply of money
Demand for Money (Liquidity Preference)
3 Motives
Transaction Motive
Precaution Motive
Speculative Motive

According to Keynes, Aggregate supply is an increasing function of level of employment (N). Aggregate supply curve will be perfectly inelastic (vertical straight line) at a point where economy attains full employment. In figure figure, at full-employment (N_f) , the AS curve becomes a vertical straight line.

Thus, ASF = f(N)

With an increase in the level of employment (N), aggregate demand tend to rise and vice versa.

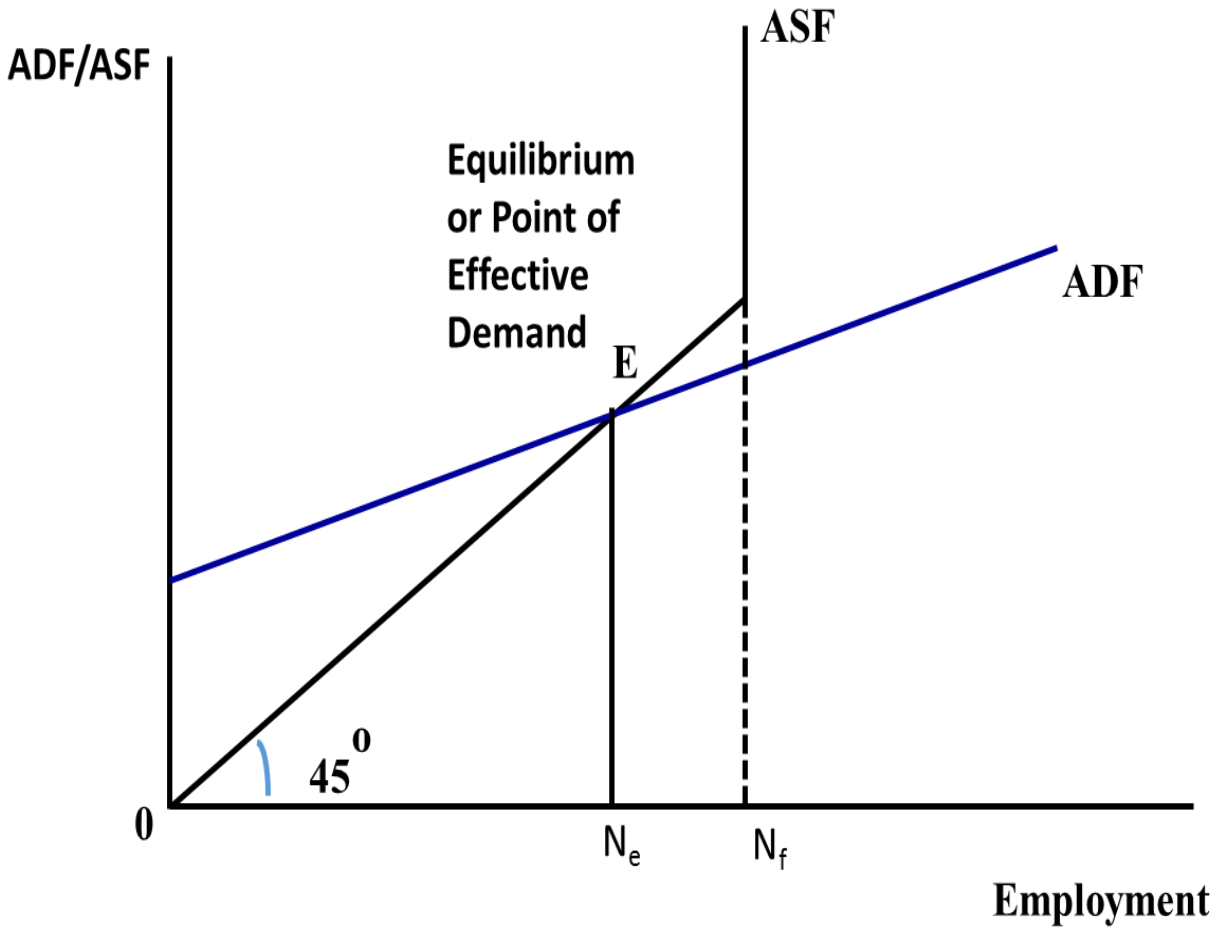
Thus, ADF = f(N)

❑ The Point of Effective Demand – Equilibrium level of Employment :

The interaction between ASF and ADF determines the level of effective demand, employment and income. So long as ADF >ASF, the entrepreneur would be induced to provide increasing employment, and this would continue till both ASF and ADF are equalised. The economy reaches equilibrium level of employment or point of effective demand when ADF =ASF.

In figure, the point of effective demand and equilibrium of the economy is attained at point E where ASF intersects with ADF. The point of effective demand (E) determines the actual or equilibrium level of employment (N_e) and output. According to Keynes, the equilibrium between ADF and ASF does not imply that the economy is necessarily having full employment at this point rather it can and often does take place at less than full employment. To him full employment equilibrium (N_f) is a rare phenomenon.

Of the two determinants of level of effective demand, Keynes assumes that ASF is given in short run. Given aggregate supply, effective demand can be raised by increasing aggregate demand.



DETERMINATION OF INCOME: ANALYSIS OF KEYNESIAN CROSS AND EFFECT OF INCREASE IN EFFECTIVE DEMAND

The equilibrium level of output, income and expenditure is determined at the point where aggregate demand (AD) is equal to aggregate supply(AS).

At equilibrium, Aggregate Demand =Aggregate Supply …….. (i)

Keynes assumes that level of aggregate supply is given in short period. Hence the level of aggregate demand determines the level effective demand and level of aggregate income . Let us assume simple two sector macro-economic model in which all savings are made by household, and there is no government spending and taxation. Thus such that aggregate demand is composed of two elements such as : consumption expenditure of households (C) and the investment decision of the firms (I).

i.e. Aggregate Demand (AD) = Consumption + Investment …….. (ii)

Further, since for every possible level of output, an equivalent amount of money income is generated. Further, income is either spent or saved.

Thus, Aggregate Output = Aggregate Supply =Aggregate Income = Consumption + Saving …….. (iii)

Further, aggregate income (= aggregate output) is in equilibrium where saving is exactly equal with investment. This is shown in lower portion of Figure-1 in which at point 'e' saving line intersect with investment line. Since saving (ae) amounts to a leakage in income spending, to maintain the flow of expenditure , an equivalent amount of investment is essential to match that leakage. It follows that saving-investment equality is fundamental condition of the equilibrium level of income which is called Keynesian cross. Thus, at equilibrium level of income , two conditions of equilibrium may be inferred

Aggregate Demand = Aggregate Supply
Saving =Investment

Keynes views that the equilibrium level of income or effective demand, the economy is not necessarily full-employment equilibrium. Usually, it can be at any point of less than full-employment level. According to Keynes, Full-employment is a rare phenomenon.

CONSUMPTION FUNCTION OR PROPENSITY TO CONSUME

Consumption function or propensity to consume studies the relationship between consumption expenditure and disposable income.

The consumption function may be written as in the following equation:

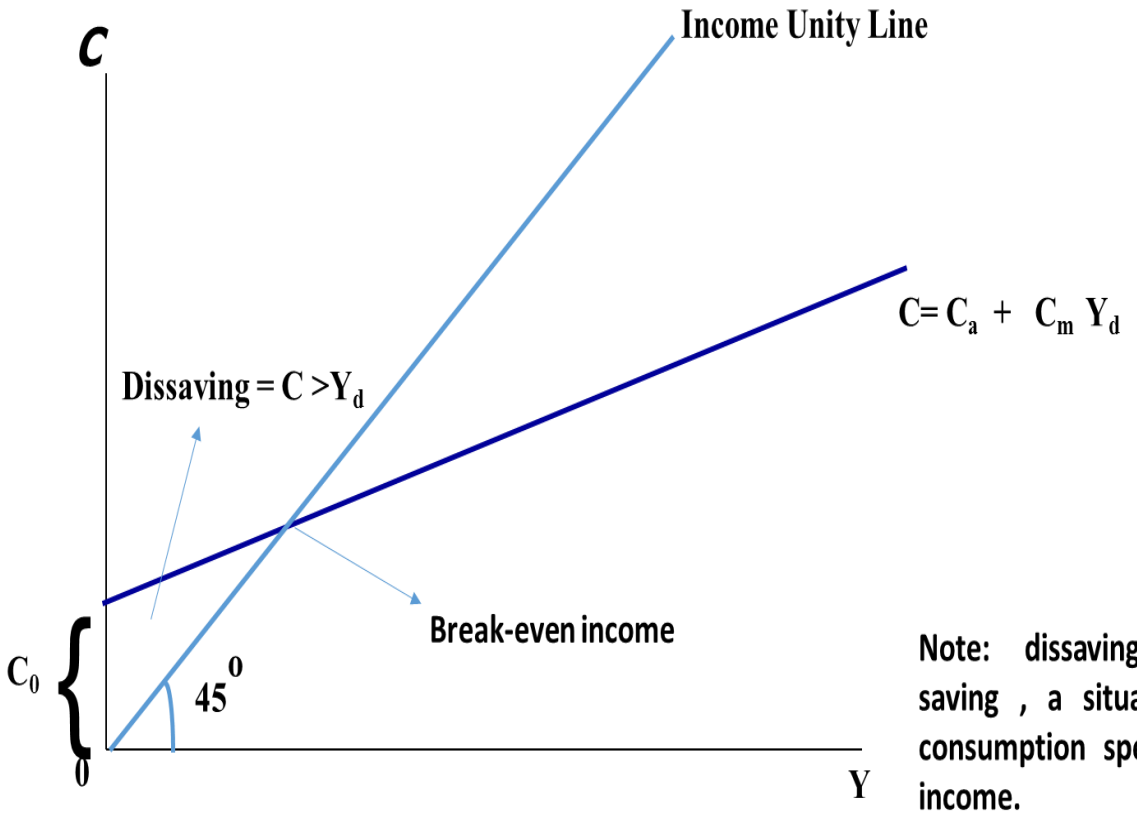
$$C = f(Y)$$

$$C = C_a + C_m Y_d$$

Where C_a = autonomous consumption

C_m = mpc (marginal propensity to consume)

Y_d = disposable income = income after deduction of tax



Note: dissaving is negative saving , a situation in which consumption spending exceeds income.

SAVING FUNCTION OR PROPENSITY TO SAVE

Keynes views that saving is the excess of disposable income over consumption. Thus saving function or propensity to save explains the relationship between saving and disposable income.

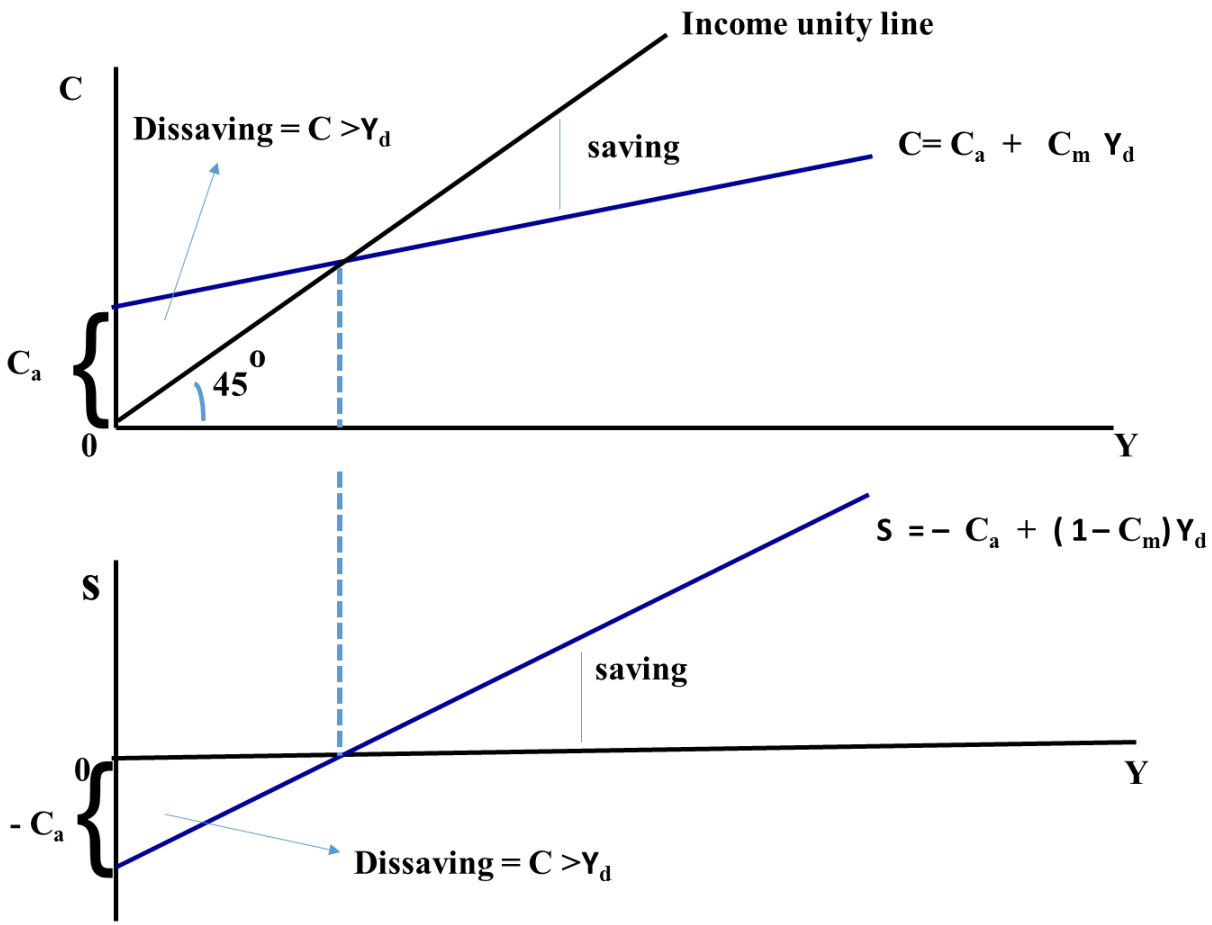
Saving function can be expresses as :

$$S = Y_d - C$$

$$= Y_d - C_a - C_m Y_d$$
$$= - C_a + Y_d - C_m Y_d$$
$$= - C_a + (1 - C_m) Y_d$$

Where, $- C_a$ = dissaving (negative saving) at zero level of income

$(1 - C_m)$ = marginal propensity to save



TECHNICAL ATTRIBUTES OF CONSUMPTION FUNCTION

- **Average Propensity to Consume :** Average Propensity to Consume (apc) is the ratio of total consumption expenditure to total disposable income.

i.e. $apc = \frac{total\ consumption\ expenditure\ (C)}{total\ disposable\ income\ (Y_d)}$

- **Marginal Propensity to Consume :** Marginal Propensity to Consume (mpc) is the ratio of change in total consumption expenditure to change total disposable income.

i.e. $mpc = \frac{change\ in\ total\ consumption\ expenditure\ (\Delta C)}{change\ total\ disposable\ income\ (\Delta Y_d)}$

TECHNICAL ATTRIBUTES OF SAVING FUNCTION

- **Average Propensity to save :** Average Propensity to save (aps) is the ratio of total saving to total disposable income.

i.e. $aps = \frac{total\ saving\ (S)}{total\ disposable\ income\ (Y_d)}$

- **Marginal Propensity to save :** Marginal Propensity to saving (mps) is the ratio of change in total saving to change total disposable income.

- i.e. $mps = \frac{change\ in\ total\ saving\ (\Delta S)}{change\ total\ disposable\ income\ (\Delta Y_d)}$

- **Relationship Between apc and aps :**

We have , We have, $Y_d = C + S$

$$or, \quad \frac{Y_d}{Y_d} = \frac{C + S}{Y_d} = \frac{C}{Y_d} + \frac{S}{Y_d} = apc + aps$$

$$or, \quad apc + aps = 1$$

- **Relationship Between mpc and mps:**

We have, $Y_d = C + S$

Differentiating with respect to Y_d we get,

$$\frac{dY_d}{dY_d} = \frac{d(C + S)}{dY_d} = \frac{dC}{dY_d} + \frac{dS}{dY_d} = mpc + mps$$

$$or, \quad mpc + mps = 1$$

INVESTMENT FUNCTION

Keyes treated investment as real or physical investment such as spending on fixed assets or fixed capital goods (like machines, equipments), inventories etc which is used for further production. From the view point of economy, investment may be treated as autonomous investment and induced investment.

Autonomous Investment : Autonomous investment is independent of change in income, rate of interest, or rate of profit. Volume of autonomous investment is fixed at different level of income, and is affected by invention or discovery of new goods, change in size of population, change in consumer's demand, research and development. Government investment expenditure are mostly autonomous in nature.

Induced investment: **Induced investment are made with a view to earn profit and are influenced by level of income, and volume of profit. Keynes highlighted that induced to invest (I) depends on two determinants such as marginal efficiency of capital (e) and rate of interest (i).**

i.e. , I = f (m.e.c. , i)

Marginal efficiency of capital (m.e.c) is the expected rate of return on capital goods , and is extremely volatile as it is affected by market optimism and pessimism in capitalist countries. Keyes views that in short-run interest rate is relatively a stable factor and does not change violently .

Given the rate of interest, m.e.c. is the most significant factor in determining inducement to invest. Private entrepreneur would be induced to invest , if there is positive gap between m.e.c. and rate of interest which in turn affect the volume of income and employment in an economy.

KEYNESIAN THEORY OF EMPLOYMENT: THE PRINCIPLE OF EFFECTIVE DEMAND

The principles of effective demand lies at the heart of Keynes's General Theory of Employment. Keynes used the term 'effective demand' to denote actual total demand for goods and services (both consumption and investment expenditure) by people in a community. The level of effective demand determines the level of employment which in turn determines the level of output and income in the economy.

❑ Factors Determining Effective Demand:

Effective demand refers to the level of demand which corresponds to equality between aggregate supply function (ADF) and aggregate demand function (ASF). Aggregate supply refers to the minimum revenue (sale proceeds) that the entrepreneurs in the economy as a whole must get from the sale of output at different level of employment. Aggregate demand represents the maximum revenues expected by the entrepreneur in the economy as a whole from the sale of output at different level of employment.

Ref: D M Mithani and Internet